

Sajeevan (Saj) Veeriah

MECHATRONICS AND ROBOTICS ENGINEER / AI AND ML / EMBEDDED SYSTEMS / END-TO-END AUTOMATION

Geelong, VIC / Open to Melbourne and broader Victoria / +61 498 586 654 / sajeevanveeriah@gmail.com
[linkedin.com/in/sajeevan-veeriah](https://www.linkedin.com/in/sajeevan-veeriah) / sajeevanveeriah.github.io / Member, Engineers Australia

SUMMARY

Mechatronics engineer specialising in **robotics, AI/ML and end-to-end automation** across mechanical design, electronics, embedded firmware, controls, software, data and system validation. Delivers integrated physical systems from requirements and prototyping through commissioning, verification and handover. Experience spans ROS 2 mobile robotics, embedded and IoT platforms, GMP smart-factory integration, automotive testing and advanced manufacturing. Combines hands-on fault finding with Python/C/C++, CAD, sensor fusion and evidence-driven technical documentation.

SKILLS	Robotics and autonomy	ROS 2 Humble, Nav2, SLAM, Gazebo, RViz, DDS, EKF and state estimation, sensor fusion, LiDAR, IMU, odometry, kinematics, PID motion control
	Embedded and electronics	C/C++, ESP32, STM32, FreeRTOS, UART, I2C, SPI, CAN, ADC, PWM, PCB design, KiCad, signal conditioning, sensors, motors, drives, actuators
	AI/ML and software	Python, MATLAB/Simulink, OpenCV, YOLO, data analysis, anomaly detection, TypeScript, REST APIs, Git, Linux, Docker, CI/CD
	Mechanical and product	SolidWorks, Fusion 360, 3D CAD, mechanism design, prototyping, 3D printing, instrumentation, engineering drawings, manufacturing support
	Automation and integration	Requirements, controls integration, PLC/HMI/SCADA, Modbus, Profinet, MQTT, LoRaWAN AU915, field devices, FAT and SAT, commissioning, root-cause analysis
	Validation and delivery	Verification and validation, test automation, CANoe and CANalyzer, ADR and EURO testing, FMEA, ISO/IEC/AS/NZS, JIRA, technical documentation, stakeholder support

EXPERIENCE	Automation Engineer	Jan 2026 to Jun 2026
	JAG Process Solutions Pty Ltd, Brunswick, VIC	
	- Delivered end-to-end automation and systems integration for pharmaceutical, biotech and food manufacturing under GMP, spanning control logic, HMI and process visualisation, field devices, drives, production data, testing and handover. - Executed an iFIX to PVI+ migration, verified functional behaviour against the validated system, and produced FAT, SAT, commissioning, qualification and handover evidence. - Supported MiR autonomous mobile robot operations and material-flow workflows in a smart-factory environment, diagnosing issues across robot, control, network and production interfaces.	
	Technical Assistant	Jul 2024 to Aug 2025
	ABMARC, South Geelong, VIC	
	- Conducted ADR and EURO vehicle emissions, EV and PHEV range and systems testing using calibrated instrumentation, data acquisition and CAN tools, analysing deviations and supporting evidence-based fault isolation. - Prepared QA, regulatory and test documentation for certification and audit, maintaining traceable configurations, results and technical findings.	
	Consultant Engineer, IoT and Projects Administrator	Feb 2024 to Aug 2024
	DuxTel Pty Ltd, Greater Geelong, VIC	
	- Designed and deployed end-to-end IoT systems linking sensors, embedded devices, gateways, Linux services, data pipelines and dashboards using ESP32, LoRaWAN, MQTT and ChirpStack, InfluxDB and Grafana. - Integrated CAN trace, GPS, custom PCB, environmental sensing and MikroTik networking for agricultural equipment monitoring, running field trials and supporting provisioning, fault finding and handover.	
	Manufacturing and Engineering Foundation Roles	2018 to 2024
	Thornton Engineering, IDL and Carbon Revolution, Geelong and Corio, VIC	
	- Supported structural-steel engineering and QA through drawing review, ITPs, MDRs, inspection planning, material traceability and AS/NZS compliance records. - Operated and monitored beverage and carbon-fibre production equipment while maintaining safety, quality, traceability and first-level troubleshooting discipline.	

PROJECTS

- **Autonomous Navigation Rover.** Developed a modular differential-drive ROS 2 Humble autonomy stack using Gazebo and RViz, Nav2 and SLAM, PID motion control, and EKF fusion of wheel odometry and IMU data for localisation and obstacle-aware navigation.
- **ESP32 Clinical Ataxia Assessment Device.** Designed and tested an ESP32-WROOM measurement prototype with four Hall-effect sensors, 100 Hz acquisition, a real-time state machine and MATLAB analysis, executing accuracy, direction, reversal, drift and temperature protocols across 17 recorded trials.
- **Agricultural Equipment Telemetry Platform.** Built a CAN and GPS monitoring workflow with a custom PCB, ESP32 sensing, LoRaWAN AU915 and MQTT telemetry, ChirpStack, MikroTik networking, Linux services, InfluxDB and Grafana dashboards, and field trials.

EDUCATION

Bachelor of Mechatronics Engineering (Honours) Deakin University, Geelong, VIC. Distinction.	2025
Higher National Diploma, Mechatronics, Robotics and Automation Engineering Cardiff Metropolitan University, United Kingdom. Distinction.	2016

DETAILS

Membership	Member, Engineers Australia
Licence	Current Victorian driver licence
Selected training	Lean Six Sigma Foundation, JIRA and Agile, KAIZEN, Industrial Automation and IIoT, AI/ML, CAD
Languages	English, Tamil, Sinhala

REFERENCES

Professional and community references available on request.